

Program: Diploma in Artificial Intelligence	
Course Code: 5389	Course Title: Virtual Reality Lab
Semester: 5	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- Gain proficiency in setting up the basic environment for Virtual Reality (VR), including understanding VR hardware, software, and system requirements.
- Learn to design and implement interactive VR scenes that enhance user experience.

Course Prerequisites:

Course code	Course name	Semester
	Basic programming skills	1-4

Course Outcomes :

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Setup the basic environment for Virtual Reality.	8	Applying
CO2	Develop VR scenes.	10	Applying
CO3	Develop interactive scenes in VR environments.	12	Applying
CO4	Develop open ended projects in VR environment	12	Applying
	Lab Exam	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3	3	3	3	3	3	3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of the Experiment	Duration (Hours)	Cognitive Level
CO1	Setup the basic environment for Virtual Reality		
M1.01	Install Unity and Visual Studio.	2	Applying
M1.02	Set up Unity for VR development.	3	Applying
M1.03	Demonstration of the working of different VR gadgets (Eg: HTC Vive, Google Cardboard, Google Daydream, Samsung gear VR etc.)	3	Applying
CO2	Develop VR scenes.		
M2.01	Develop a scene in Unity that includes: cube, plane and sphere.	2	Applying
M2.02	Apply transformations on game objects (cube, plane etc.).	2	Applying
M2.03	Add multimedia sources to a scene.	2	Applying
M2.04	Create new materials for game objects and change their color and texture separately.	2	Applying
M2.05	Apply Rigid body component, material and Box collider to the game Objects.	2	Applying
	Lab Exam – I	1.5	

CO3	Develop interactive scenes in VR environments.		
M3.01	Develop a scene in Unity that includes different Game objects. Write a C# program to change the colour and material/texture of the game objects dynamically on button click.	4	Applying
M3.02	Develop a scene in Unity that includes a sphere and plane. Apply Rigid body component, material and Box collider to the game Objects. Write a C# program to grab and throw the sphere using a VR controller (or simulator).	4	Applying
M3.03	Develop a simple UI (User interface) menu with images, canvas, sprites and buttons. Write a C# program to interact with UI menu through VR trigger button such that on each successful trigger interaction displays a score on scene.	4	Applying
CO4	Develop open ended projects in VR environment		
M4.01	Open Ended Experiments**	12	Applying
	Lab Exam – II	1.5	

**** - Suggested Open Ended Experiments**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open ended experiments as a group of 2-3. There is no duplication in experiments between groups.)

1. Golf Game
2. flying helicopter
3. moving car
4. architecture of a house

Programming tools recommended: Unity, C#, Blender, VRTK.

Text / Reference

T/R	Book Title/Author
T1	Virtual Reality Systems, John Vince, Pearson Education Asia, 2007
R1	Developing Virtual Reality Applications, Alan Craig, William Sherman and Jeffrey Will, Foundations of Effective Design, Morgan Kaufmann, 2009.

Online Resources

Sl.No	Website Link
1	https://learn.unity.com/
2	https://www.coursera.org/specializations/programming-unity-game-development